# Agent Task Contract
# SME Procurement-Preparation Agent
# Week 5 Deliverable BSE4104

**Agent Goal**

*Primary Goal*

Given a user request about one or more inventory items, the agent shall autonomously,

- Check the current inventory status of the relevant item.
- Determine whether a reorder is required.
- If a reorder is required, retrieve available supplier quotations.
- Create a draft purchase requisition with appropriate quantity and justification.
- Stop and hand over the draft to a human for approval.

The agent must never approve the requisition, select a final supplier, place an order, or commit funds.

**Allowed Tools**

The agent may only use the following approved tools

| Tool name | Purpose | Side effect | Permission |
|---|---|---|---|
| get_inventory | Retrieve stock level and reorder information | None | Allowed |
| get_quotations | Retrieve supplier quotations for an item | None | Allowed |
| create_draft_requisition | Create a purchase requisition in Draft status | Creates draft record | Allowed |

*Explicitly Forbidden Actions*

- Approving a requisition
- Changing status to “Approved”
- Placing an order
- Sending documents to suppliers
- Committing budget or making payments
- Any tool not listed above

**Agent State**
The agent shall maintain the following minimal state during a run:

- current_item_id
- inventory_status
- quotations
- draft_requisition_id
- steps_taken
- status of one of: running, completed, stopped_human_handoff, stopped_limit_reached, error

State is session-scoped and does not persist across different user conversations unless explicitly saved as a case.

**Limits and Constraints**

| **Constraints** | **Value** | **Rationale** |
|---|---|---|
| Maximum iterations / turns | 8 | Prevents infinite loops |
| Maximum tool calls per run | 6 | Keeps behaviour predictable and cost-controlled |
| Temperature | 0.2 | Favours consistent, low-creativity behaviour |
| Tool choice | auto | Model can decide order of tools |

**Stop Conditions**
The agent must stop and return control when any of the following occur:

*Success Stop*
- A draft purchase requisition has been successfully created and the user has been informed that it is awaiting human approval.

*No Action Needed*
- Inventory check shows that the item is above the reorder point (no requisition required).

*Human Hand-off Required*
- Missing or invalid data that the agent cannot resolve
- Ambiguous user request
- Any situation requiring judgement beyond the approved tools

*Safety*

- Maximum iterations reached
- Maximum tool calls reached
- Repeated tool errors

*Error Stop*

- Unrecoverable tool failure or unexpected system error.

**Human and Approval Conditions**

Human involvement is mandatory in the following cases,

- Final approval or rejection of any draft requisition
- Selection of a final supplier
- Any decision that commits money or creates a legal obligation
- When the agent explicitly returns status stopped_human_handoff
- When the user requests an action outside the agent's allowed tools

The agent must clearly communicate when it is handing over to a human and what decision is required.

**Success Criteria for a Single Run**

A run is considered successful when,

- The agent used only approved tools
- It correctly determined whether a reorder was needed
- If needed, it created a draft requisition with valid data
- It stopped cleanly and informed the user of the next required human action
- All actions are fully traceable in the execution log